CASE STUDY

One security, two factor panels

What the model says about Amazon.com — and which version of it you should believe

Bastian Offermann · September 2026 · US:AMZN · 6,671 trading days

The short version. Estimated on identical settings, the **raw** panel fits Amazon better in sample — adjusted R^2 of 0.452 against 0.418 — and forecasts its risk distinctly worse out of it. Its Mincer-Zarnowitz slope is 0.445 against 0.614, and where the orthogonalised forecast survives the joint test (p = 0.405) the raw one does not (p = 0.008). It also throws away 2 windows as unusable, against 0. The orthogonalised panel is the one to forecast with. The raw panel is the one to explain an exposure with, and this document shows exactly where the two disagree.

1 The security

Amazon is a deliberately awkward test case. Over 6,671 trading days from 2000-01-04 it compounded at 16.4% a year at 48.0% annualised volatility, with a worst day of -28.5% and a best of 29.6%, and an excess kurtosis of 13.1. It changed what it was twice — online retailer, then cloud infrastructure business — without changing its ticker, which is precisely the situation a return-based model is built for and a holdings-based one struggles with.

1.1 The experiment

Two specifications, identical in every parameter except the factor panel they read.

	Orthogonalised	Raw
Specification	`88b9b8e7d5af...`	`a8d1454d32ea...`
Factor panel	block-hierarchy residuals	excess returns, before residualisation
Window / step	252 days, rolled monthly	252 days, rolled monthly
Estimator	OLS, Newey-West HAC	OLS, Newey-West HAC
Covariance	blend (EWMA + shrunk)	blend (EWMA + shrunk)
Forecast horizon	21 days	21 days
Windows estimated	260	260
Forecasts scored	248	246
Sample	from 2005-10-28	from 2005-12-29

Everything that follows is a difference between these two columns, and the only thing that differs is which version of the same forty factors the regression sees.

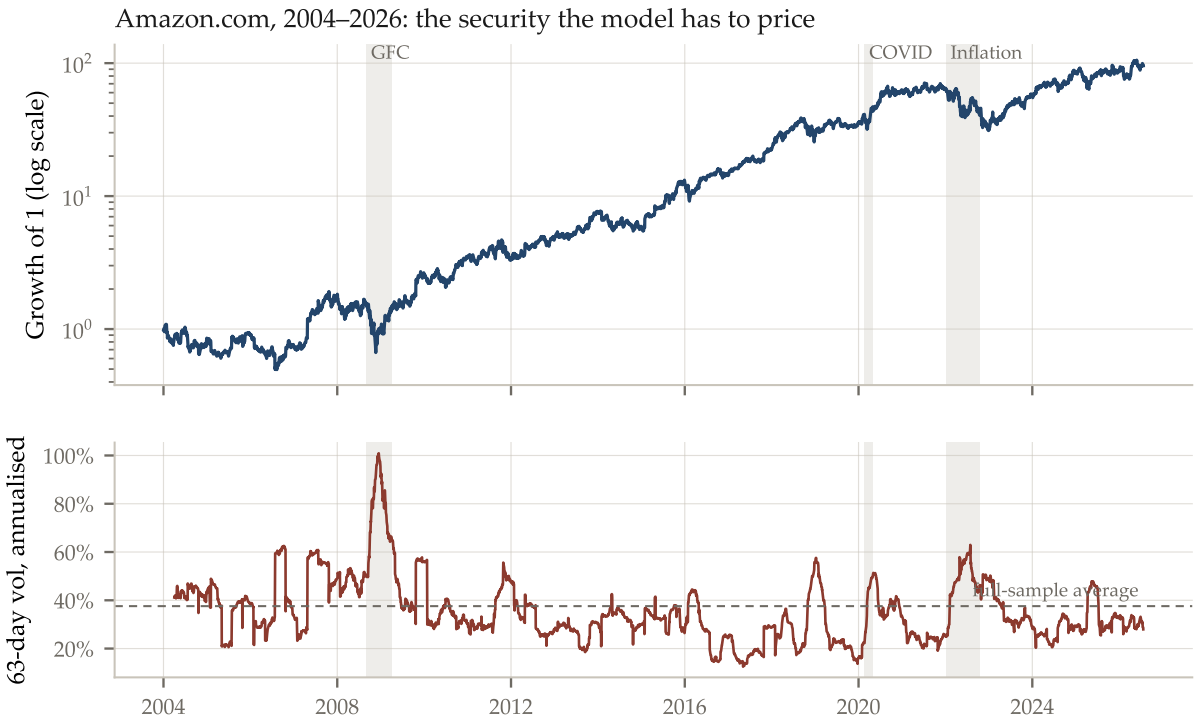


Figure 1. The security, before any model touches it. Shaded bands mark the global financial crisis, the COVID drawdown and the 2022 inflation shock; they are editorial context drawn on the chart, never an output of the model.

2 What the model says Amazon *is*

Read the equity rows first, because that is where the panels are supposed to disagree and where the disagreement is largest.

On the **orthogonalised** panel, Amazon loads 1.45 on `eq_global` and 0.36 on `eq_us`. That is the honest reading of a hierarchy: global equity carries the market exposure, and the US residual — what US equity does *beyond* global equity — adds almost nothing on top. One number does the work.

On the **raw** panel the same security loads 1.85 on `eq_global` *and* 1.76 on `eq_us`. Both are large, both are positive, and they are describing the same exposure twice, because on the raw panel `eq_us` and `eq_global` are near-copies of each other. The variance inflation factors say so out loud: 172 and 75 on the raw panel, against 11 and 8 on the orthogonalised one.

> Neither reading is wrong, and they answer different questions. *"How much equity risk does Amazon carry?"* is the raw panel's question, and the answer is the sum. *"Which part of the equity complex explains Amazon?"* is the orthogonalised panel's, and the answer is: global beta, plus a cyclical tilt, minus momentum. Only the second can be decomposed without double-counting.

The non-equity rows are more interesting than they look. `sty_momentum` comes out at -0.65 on both panels — a genuinely negative momentum loading, which for a stock that has compounded at 16.4% a year is counter-intuitive until you remember what the factor is: a long-only momentum fund residualised against market and sector. Amazon is a large, widely held growth name, not a momentum name, and the model is saying so consistently on both panels. Agreement across panels is itself evidence: a loading that survives residualisation is not an artefact of the design.

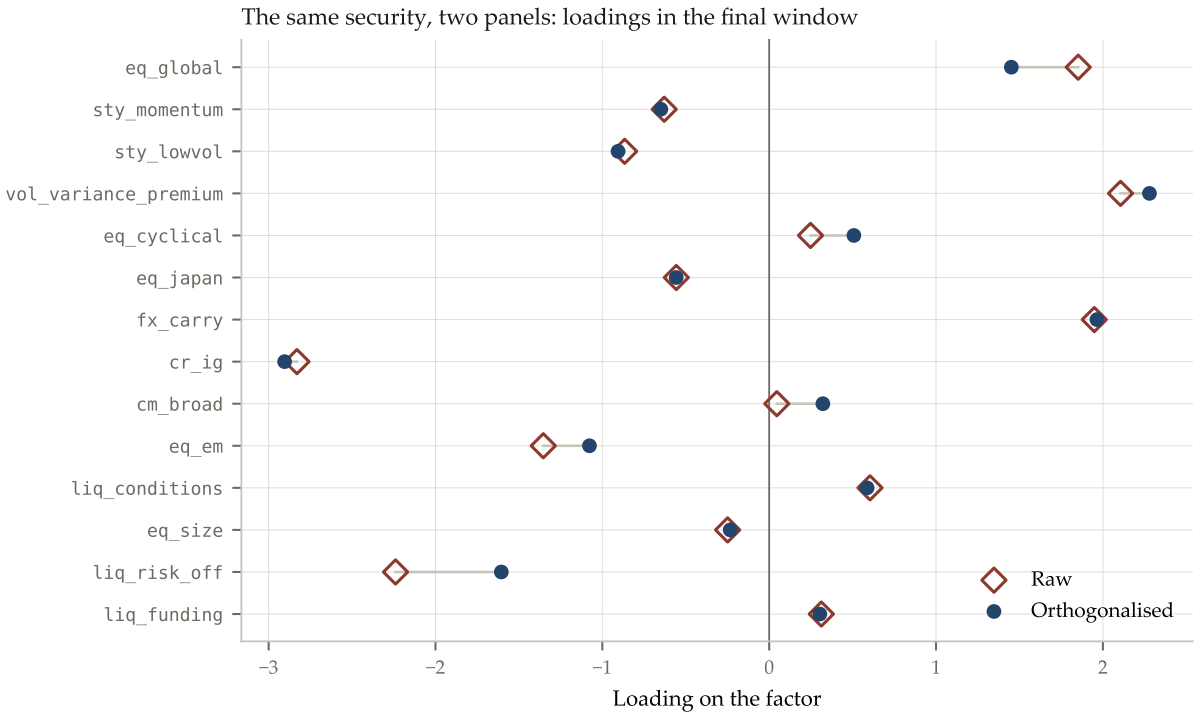


Figure 2. Loadings in the final window (2026-07-22), the fourteen largest by orthogonalised *t*-statistic. Where the two markers sit on top of one another the panels agree; where they separate, the raw loading is absorbing something the hierarchy has already assigned elsewhere.

3 Stability: the same exposure, month after month

Start with the top panel, which is the cleanest single picture in this document. Amazon's loading on `eq_global` has a standard deviation of 0.37 on the orthogonalised panel and never leaves the range 0.46 to 2.41. The same loading on the raw panel has a standard deviation of 1.93 and spans -1.74 to 6.98 — from a short position in global equity to a beta of seven, for a security whose equity exposure plainly did neither.

The two raw equity betas are negatively correlated at -0.38, which is the offsetting signature of near-collinear regressors: when one swings up the other is pulled down, and part of the movement cancels in the sum. Only part of it. The individual numbers are not identified, and an analyst reading either one alone is reading which way the collinearity broke in that window.

Aggregated over the whole vector, the raw panel's exposures move a median 7.9 per month against the orthogonalised panel's 5.5 — roughly *forty per cent further*, every month, for a security whose underlying business did not change on that schedule. The correlation with the previous window tells the same story more gently: 0.906 orthogonalised, 0.886 raw.

> An exposure estimate that moves this much between windows is not a measurement of Amazon. It is a measurement of which way the collinearity happened to break in the last 252 days.

4 Fit: where the raw panel wins

Mean adjusted R^2 is 0.452 on the raw panel against 0.418 on the orthogonalised one, and mean annualised residual volatility is 26.6% against 27.4%. The raw panel fits better. It is important to say so plainly, because it is the one statistic an analyst is most likely to look at, and on its own it points the wrong way.

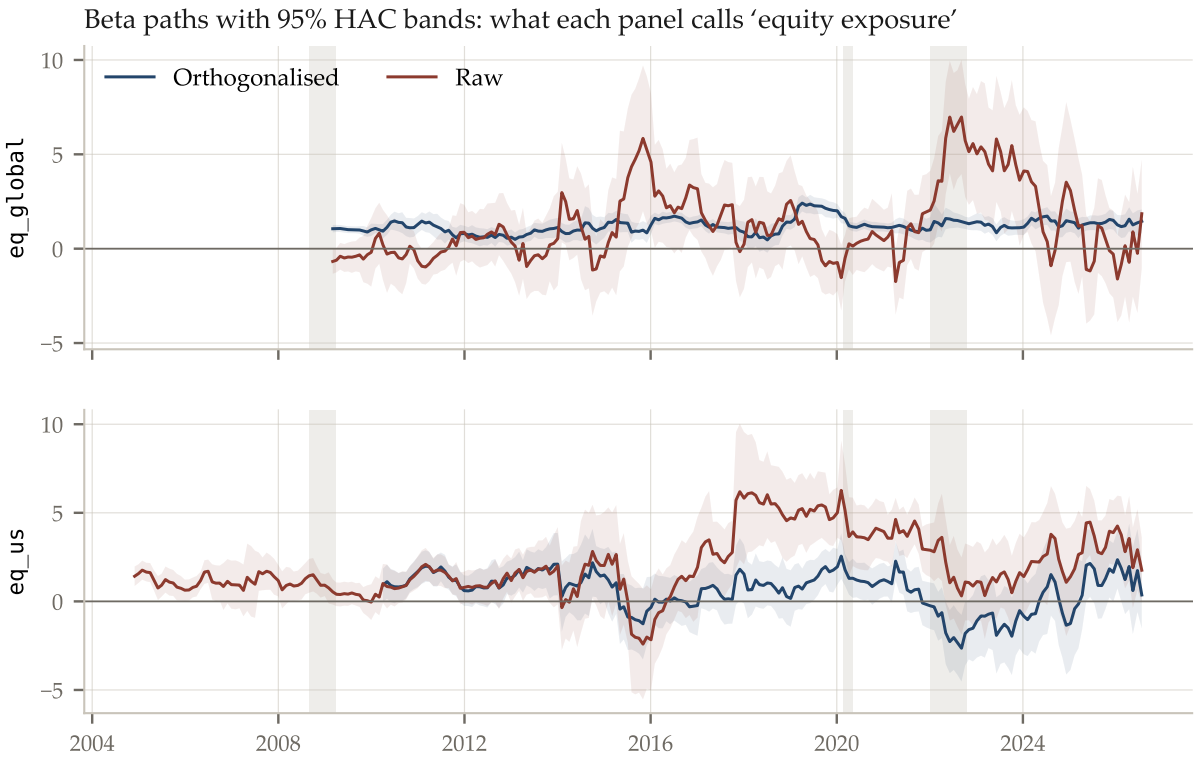


Figure 3. Equity beta through time with 95% HAC bands. Top: global equity. Bottom: US equity. The orthogonalised global-equity beta is the flat navy line in the upper panel; everything else on the chart is the same exposure, estimated less stably.

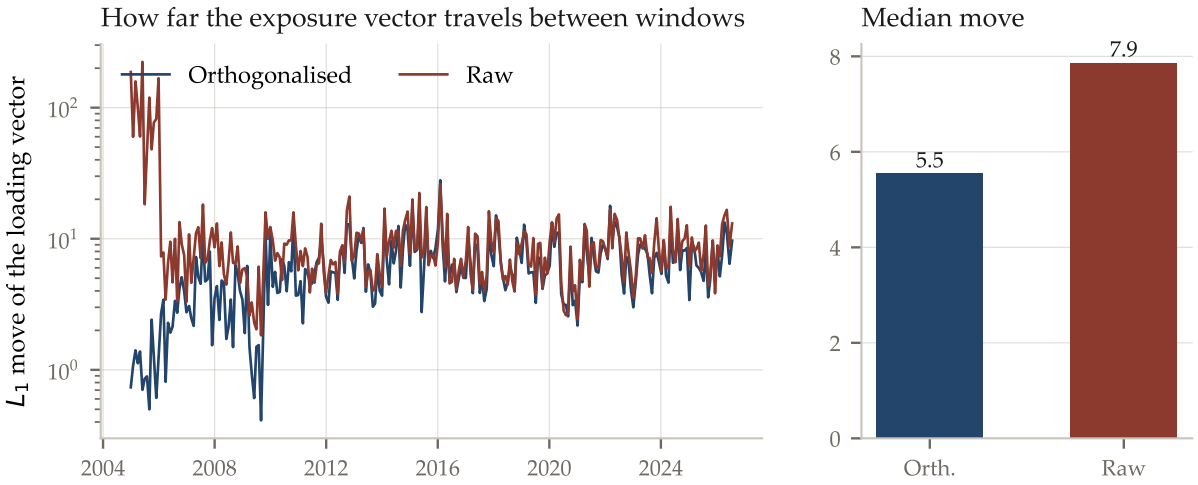


Figure 4. How far the whole loading vector travels between consecutive windows, measured in L_1 on the factors common to both. Log scale on the left.

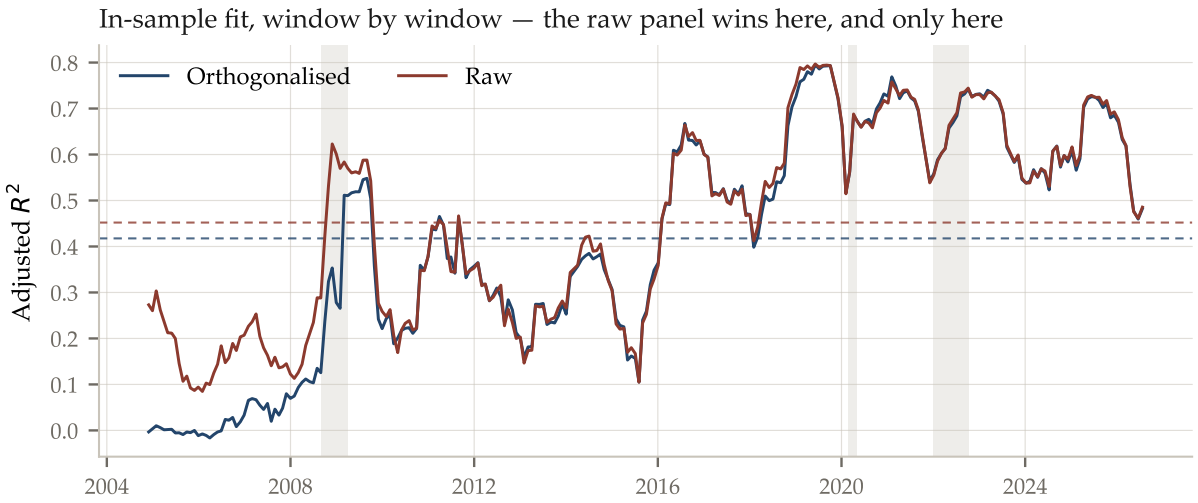


Figure 5. Adjusted R^2 window by window, with each panel's mean as a dashed line. The raw panel is above the orthogonalised one almost everywhere.

The reason is mechanical rather than economic. The residualised factors span the same space as the raw ones up to a rotation, so a perfectly conditioned regression would give identical fits. The gap is what near-collinearity buys in sample: extra freedom to fit the particular 252 days in the window, paid for with parameters that do not generalise.

The conditioning chart is the same information from the other side. The orthogonalised design averages a condition number of 16 and never exceeds 40. The raw design averages 117 and reaches 1,451. Its largest VIF averages 4,975 and peaks at 230,726 — a factor whose R^2 against the other thirty-nine rounds to 0.999996.

> In-sample fit is not evidence about a forecasting model. On this security it is evidence pointing the wrong way, and the next section is what settles the question.

5 Forecast: where the raw panel loses

Both panels track the broad shape of Amazon's volatility. The difference is in how much of the *variation* each one captures, which is what the Mincer-Zarnowitz regression measures.

	Orthogonalised	Raw	Target
Forecasts scored	248	246	—
Windows excluded as ill-conditioned	0	2	0
Mean bias ratio	0.897	0.878	1
Bias statistic (sd of standardised returns)	0.983	0.963	1
Mincer-Zarnowitz slope	**0.614**	0.445	1
Mincer-Zarnowitz joint p	**0.405**	0.008	> 0.05
Mincer-Zarnowitz R^2	0.113	0.123	high

The orthogonalised forecast has a slope of 0.614 and is *not* rejected by the joint test of $(\alpha, \beta) = (0, 1)$ at $p = 0.405$. The raw forecast has a slope of 0.445 and is rejected at $p = 0.008$. Both get the average level roughly right — bias ratios of 0.897 and 0.878 — and the orthogonalised panel's bias statistic of 0.983 is almost exactly the 1 it is aiming at. What separates them is period-by-period accuracy, and that is the property a risk limit depends on.

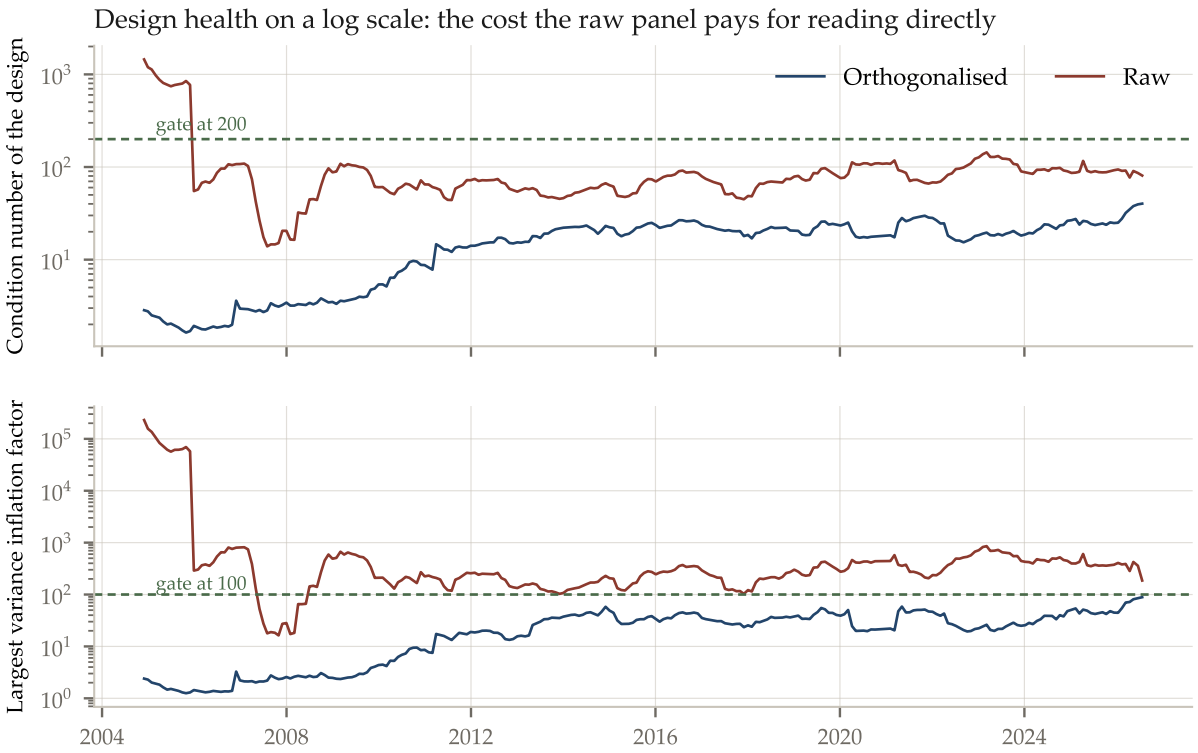


Figure 6. Design health on a log scale. The raw panel's largest variance inflation factor sits an order of magnitude above the gate for essentially the entire sample — which is why the VIF limit is applied only to the orthogonalised panel, and the condition number to both.

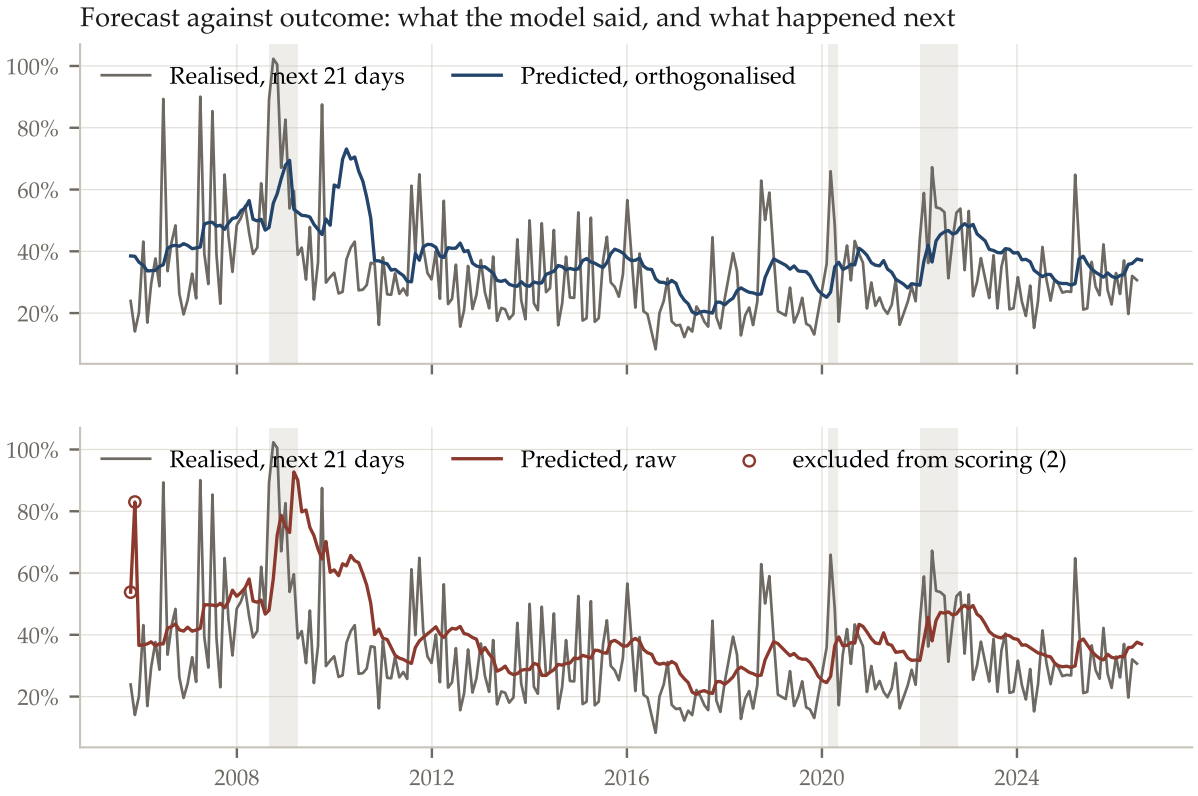


Figure 7. Each panel's 21-day volatility forecast against what actually happened over the following 21 days. Circled points on the lower chart are the 2 windows the reliability gate excluded from scoring.

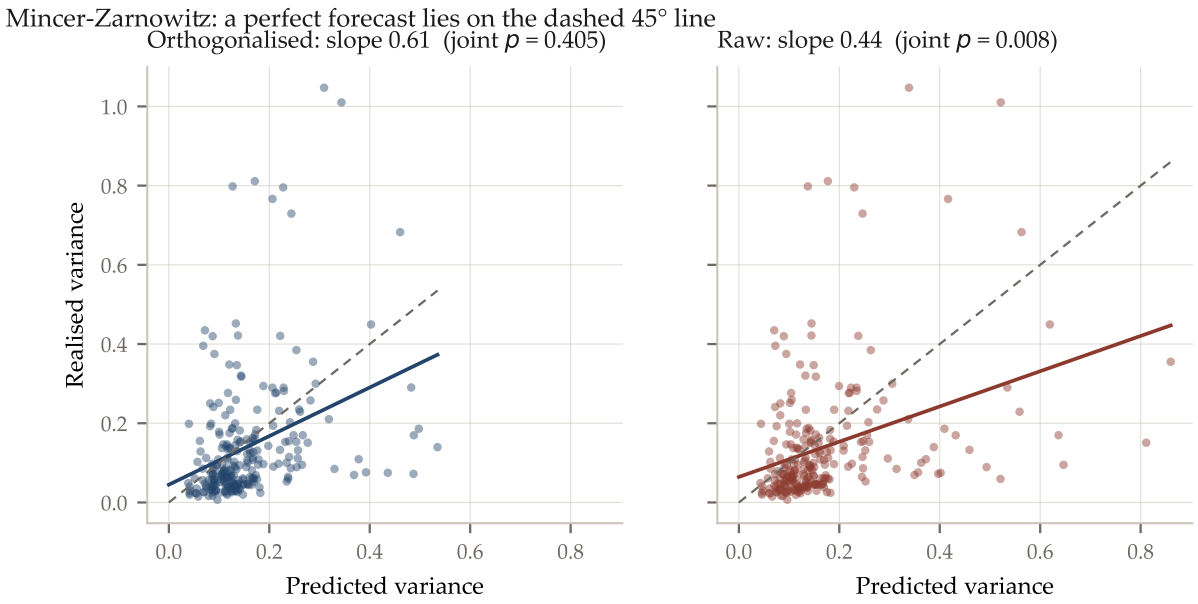


Figure 8. Realised variance against predicted variance. A perfect forecast lies on the dashed 45° line; the solid line is the fitted Mincer-Zarnowitz regression, with HAC standard errors because the forward windows overlap.

A slope below 1 is partly mechanical at a 21-day horizon, because realised volatility over a forward window blends regimes while the forecast refers to one; on simulated data a *perfect* forecast produces a slope near 0.5 when the true regime length matches the horizon. So 0.614 is a respectable number in absolute terms, and the comparison between 0.614 and 0.445 is the part that carries no caveat, because the mechanical effect is identical on both.

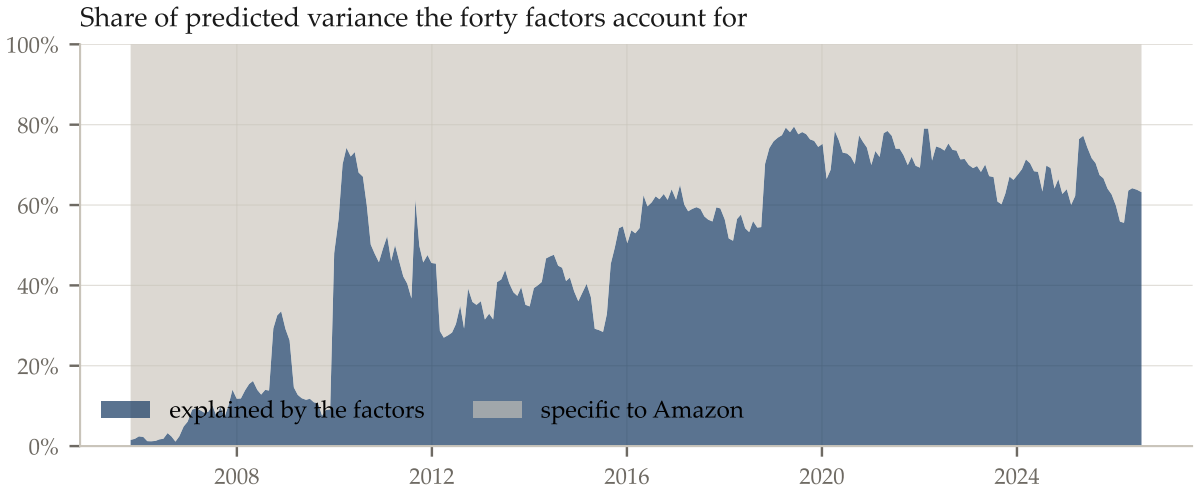


Figure 9. The share of predicted variance the forty factors account for, on the orthogonalised panel. The remainder is specific to Amazon. The factor share averages 49% and exceeds half in 57% of windows — low for a large-capitalisation equity, and the point of the chart.

6 Where the risk sits

At the final window the orthogonalised model predicts 37.1% annualised volatility for Amazon, of which 63% of the *variance* is systematic. In the Euler decomposition — which splits the volatility rather than the variance, and so sums exactly to 37.1% — the largest single factor block is Equity Market at 29%. An almost equal share, 36%, is specific: Amazon's own risk,

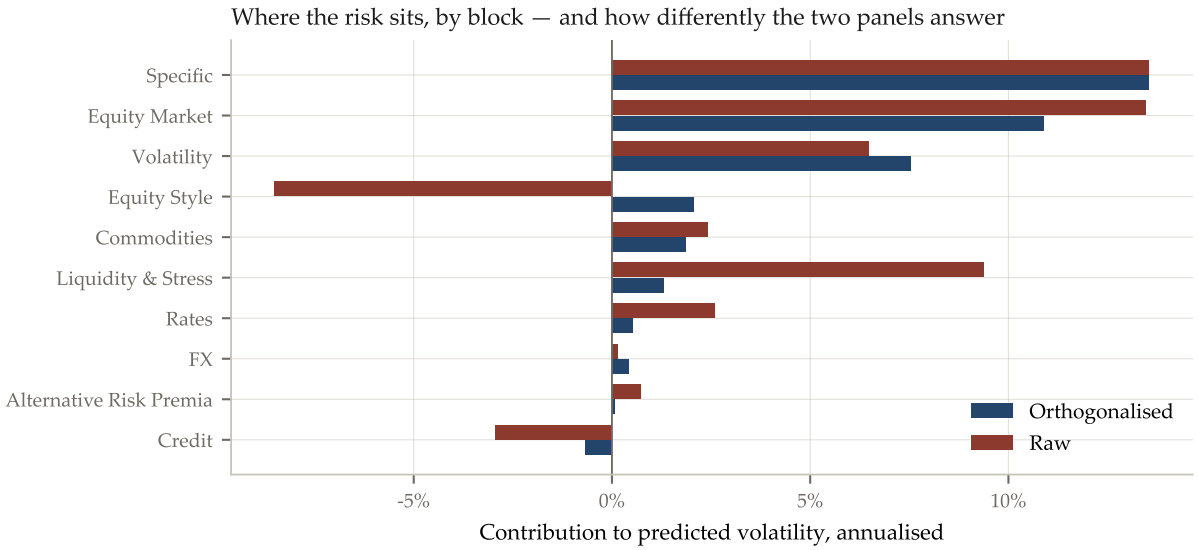


Figure 10. Euler risk contributions at 2026-07-22, aggregated to block, from the same 504-day blended covariance the forecast used. Contributions sum to the predicted volatility.

which no factor in the panel explains.

That last number is the honest headline of the whole exercise. Across the whole sample the forty factors account for an average 49% of Amazon's predicted variance, and they exceed half of it in 57% of windows. The remainder is Amazon: its margins, its capital expenditure cycle, its regulatory exposure, its cloud business. The model does not claim otherwise, and the specific-risk floor exists so that it cannot.

> The raw panel puts 36% of predicted volatility in Equity Market against the orthogonalised panel's 29% — and, look at the chart, assigns *Liquidity & Stress* several times more risk while giving *Equity Style* a large negative contribution. Same security, same day, same covariance method. The blocks disagree because on the raw panel they overlap, so each is partly crediting the others' exposure.

7 Where both panels fail

A case study that only reported the comparison it set out to make would be misrepresenting the model. Both panels fail the value-at-risk tests, and they fail in the same interesting way.

	Orthogonalised		Raw	
	95%	**99%**	**95%**	**99%**
Exceptions expected	260	52	258	52
Exceptions observed	179	75	171	71
Kupiec p	<0.001	0.003	<0.001	0.010

Look at the direction of the miss. At the 95% level there are *too few* breaches — 179 against 260 expected. At the 99% level there are *too many* — 75 against 52. The model is simultaneously too cautious in the body of the distribution and too confident in its tail.

That is not a calibration error in the volatility forecast, which the bias statistic of 0.983 says is very nearly right. It is the Gaussian quantile assumption failing. After dividing every return

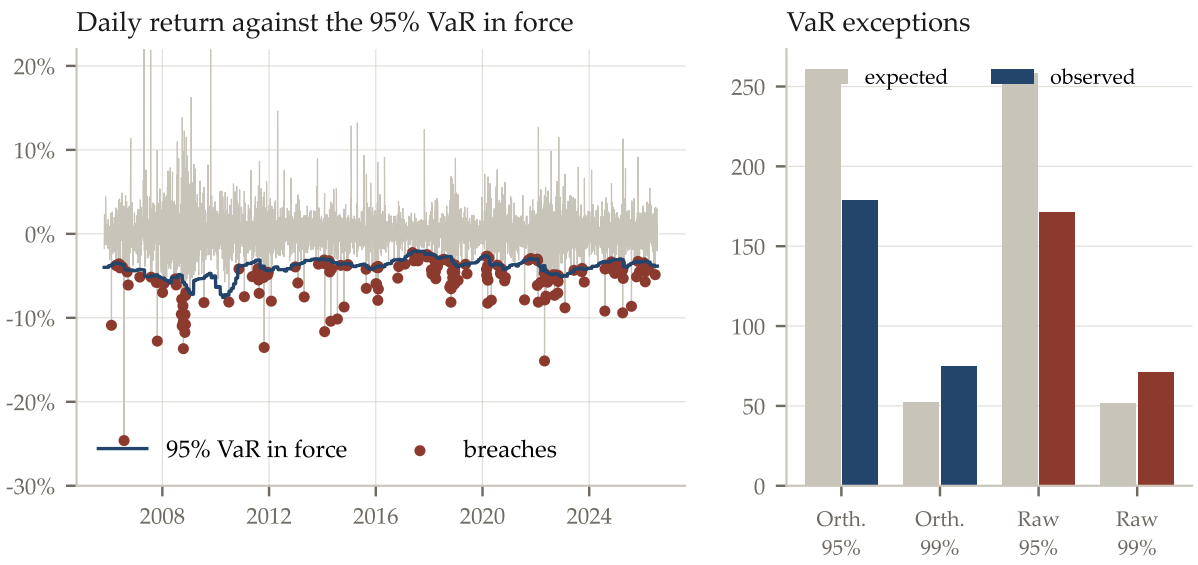


Figure 11. Left: Amazon's daily return against the 95% value-at-risk in force over the scored period, on the orthogonalised panel; the VaR steps because the forecast is refreshed monthly. Right: expected against observed exceptions at both levels and on both panels.

by the volatility forecast that was current on that day, the standardised series still has an excess kurtosis of 12.1 — barely reduced from the raw 13.1. Scaling the distribution correctly does not make it normal, and Amazon's is emphatically not.

Both panels also fail the Christoffersen conditional-coverage test, with p-values below 0.001: the breaches cluster. The cause is a forecast held for a month while volatility moves within it — the methodology review measured that re-estimating the covariance daily, with the betas held, removes most of the clustering. It is the same on both panels and therefore not evidence about the panel choice, but it is a limit of this specification and is reported as one; the model agent searches the daily refresh as an option.

The practical consequence is narrow and worth stating: use this model's volatility forecast, and treat its Gaussian VaR as a lower bound on the tail. The Student-t quantile the risk module also offers is the better default for a security like this one.

8 Verdict

Question	Answer for Amazon
How much equity risk?	Raw panel. Read `eq_global` + `eq_us` as a single total exposure; the split between them is not identified.
What explains the returns?	Orthogonalised panel. Global beta, a cyclical tilt, a negative momentum loading — each incremental over the blocks above it.
How risky is it next month?	Orthogonalised panel, 37.1% annualised. The raw panel's forecast is rejected by the joint Mincer-Zarnowitz test.
Where does the risk come from?	Orthogonalised panel. 36% of predicted volatility is specific to Amazon and no factor in the panel explains it.
How bad can a day get?	Neither, on a Gaussian quantile. The standardised returns still have excess kurtosis of 12.1.

The general finding this case study supports is the one the model is built around: the in-sample statistic and the out-of-sample statistic disagree, and the out-of-sample one is right. On a single security, with every other setting held fixed, the raw panel bought $0.452-0.418$ of adjusted R^2 and paid for it with a Mincer-Zarnowitz slope of 0.445 instead of 0.614, 2 unusable windows instead of 0, and a loading vector that travels forty per cent further every month.

Reproduce with:

```
python -m backend.pipeline.sync -security US:AMZN
python -m backend.pipeline.run_estimation -instrument US:AMZN -window 252 -step 1m
python -m backend.pipeline.run_estimation -instrument US:AMZN -window 252 -step 1m -raw-factors
python -m backend.pipeline.run_risk -instrument US:AMZN -spec 88b9b8e7d5af...  -horizon
21
python -m scripts.build_case_study -pdf
```

Every figure and every number above is generated by the last command from the database; none is typed by hand.